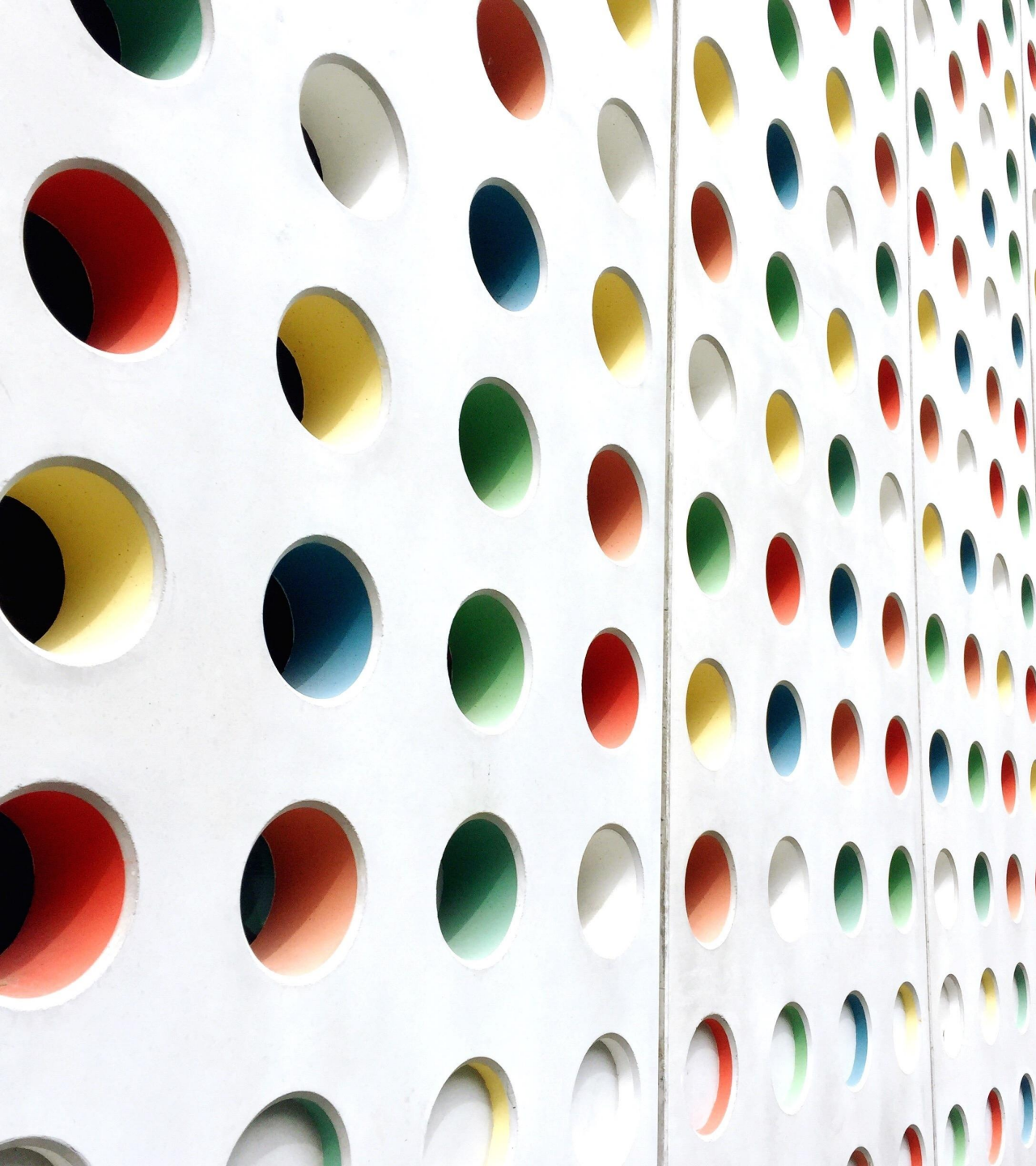




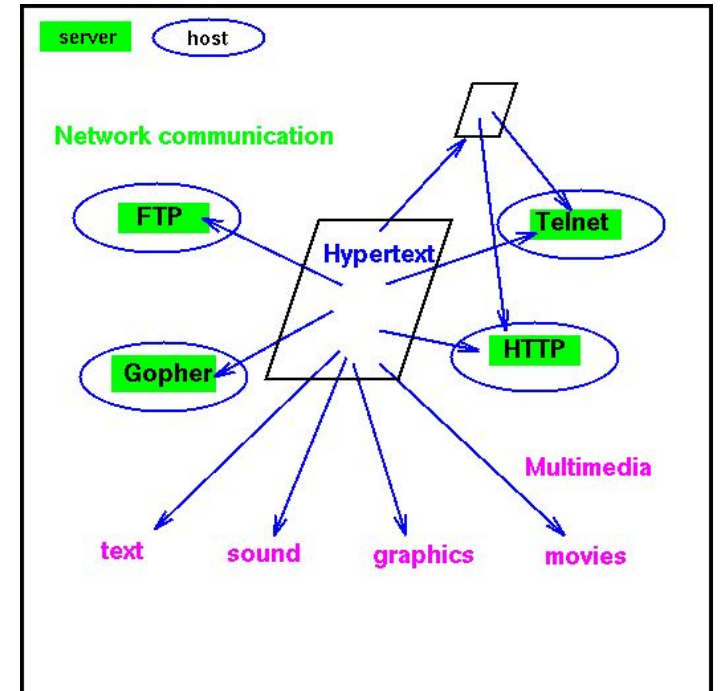
Web of hypertext: RDFa and microformats

BY

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Hypertext

- Obvious and major sources of data on the Web is hypertext--HTML and XHTML.
- HTML is not only used by individual authors to produce Web pages directly; in fact, most of the HTML pages visible through browsers are produced by various types of back-end processes.
- The information displayed through HTML, is reflection of database content.
- HTML is a natural vehicle to incorporate the data that can be processed by the Semantic Web.

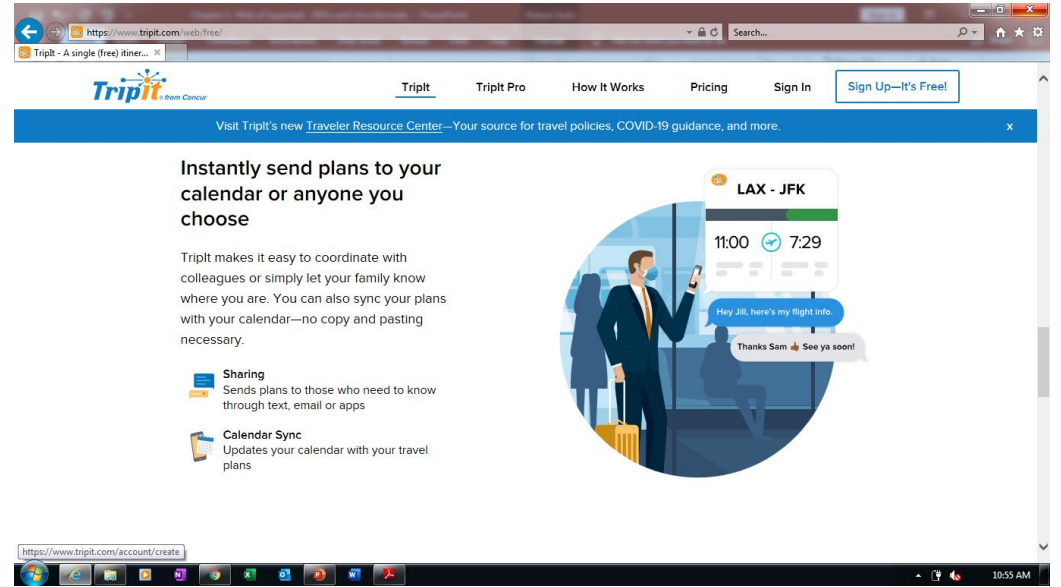
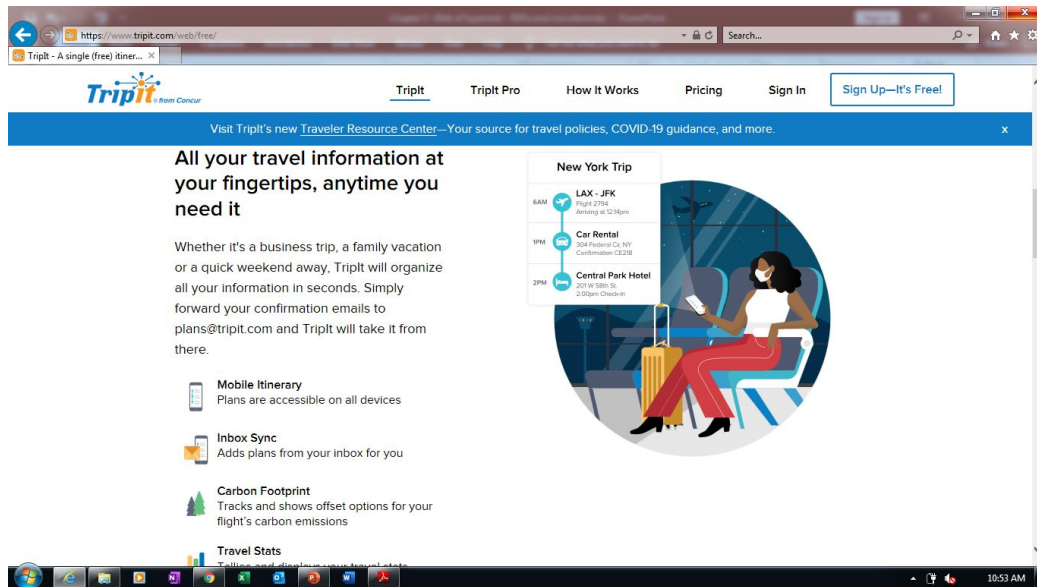
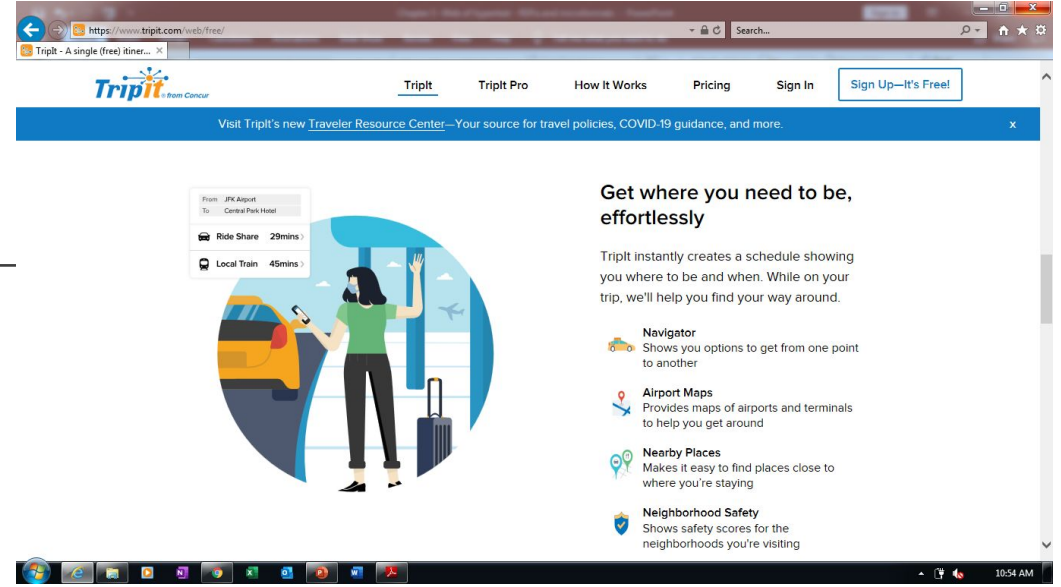
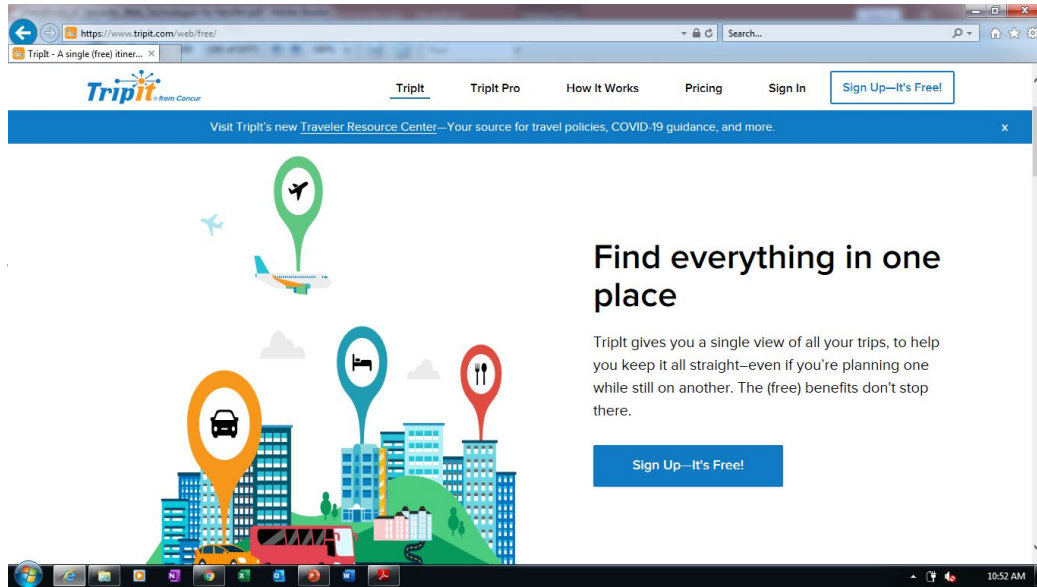




Requirement of Additional Structure

Conveniently, the (traditional) Web community has come to the realization that it is beneficial to expose more of the underlying structured information than what can simply be displayed by a browser for, essentially, human consumption.

The Web 2.0 phenomenon brought a more interactive and participative Web to the fore that often relies on additional structure within the published hypertext data.



Semantic and Traditional Web [Dual evolution]

- Dual evolution (on the Semantic and on the more “traditional” Web) saw the emergence of a common approach (though with different techniques), namely that of (X)HTML pages including additional information that provide structural, “meta” information about what is being displayed visually.
- This additional information is added to the (X)HTML content as elements or attributes, providing additional information to the text.
- Traditional Web community has developed the Microformats approach for reusing existing HTML attributes, for example, @class and @title and elements (e.g., abbr).
- Microformat vocabularies are defined and maintained by the community; vocabularies (i.e., conventions on which elements and attributes are used and for what purpose) cover different application areas.
- Applications that understand a specific microformat vocabulary and know that a specific (X)HTML page abides to that vocabulary can make use of this inherent structure in the markup to get at the raw data without screen-scraping.

Microformats

- **XHTML** files with microformat content into RDF. The transformation engine XSLT that converts the XHTML structure into RDF/XML. A W3C standard, called **GRDDL**, has been defined to enable generic processors to locate the XSLT transformation for a particular Microformat, given an appropriate declaration in the XHTML content header.
- RDFa, provides a more general microformat. RDFa relies on attributes, **RDFa** defines its own set of attributes that are used alongside existing **XHTML** constructs. RDFa provides the flexibility to add essentially any information to an XHTML page, provided that a suitable vocabulary is defined that gives sense to the URIs employed by the user.

Microformat Mapping

- RDFa specification defines a precise mapping from an XHTML+RDFa file to RDF.
- In contrast to the Microformat+GRDDL approach, there is no need to define a separate transformation engine per vocabulary; instead, one such engine can cover all usages of XHTML+RDFa, hence the suitability of RDFa to wide-scale usage.
- In fact, RDFa+XHTML may be viewed as another RDF serialization format, alongside RDF/XML or Turtle, one that provides both human and machine readability.

1. Applicability of Technologies

GRDDL and RDFa will be presented only in conjunction with (X)HTML.

Technologies can also be used for other XML dialects.

1. GRDDL-- a generic processor could find a suitable XSLT transformation for any XML schema or namespaces.
2. RDFa attribute set can be added to other XML dialects--already been done for SVG Tiny 1.2 or Yahoo!'s MediaRSS.

2. Microformats and Semantic Web

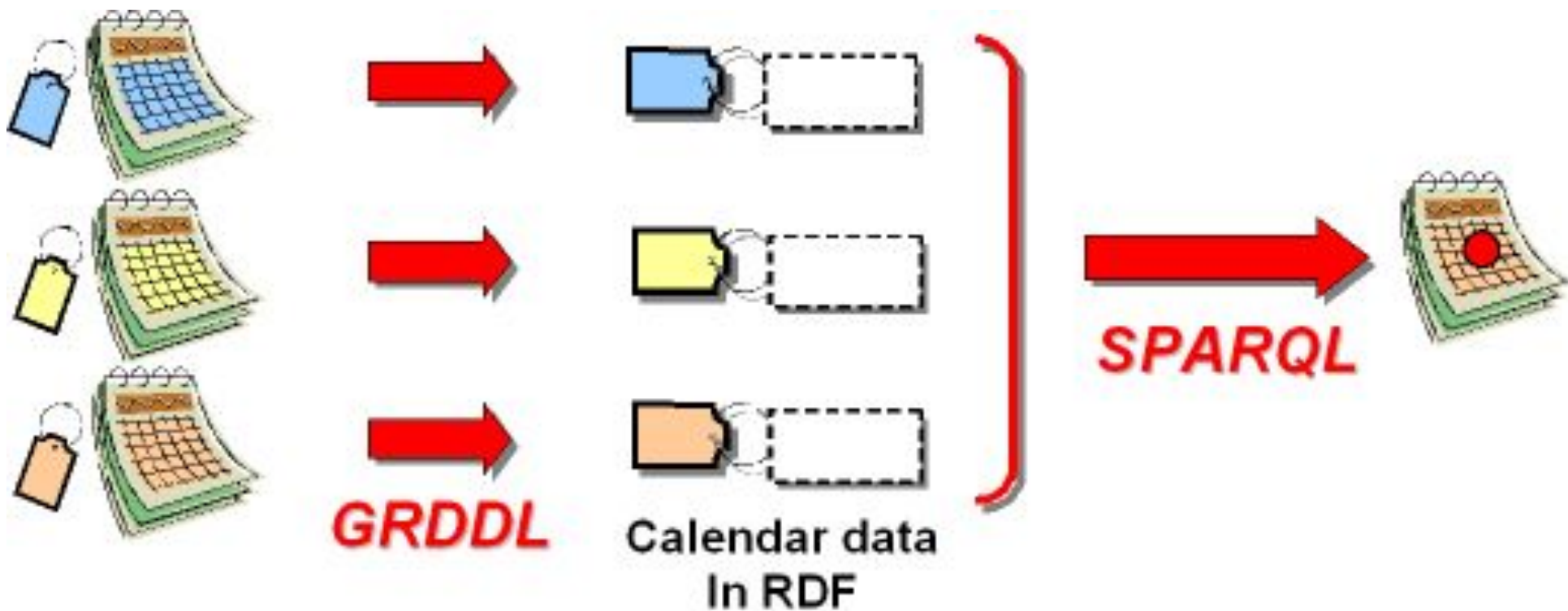
- Microformats reuse existing attributes in the HTML specification to add semantic information to the content.
- “Microformats” does not refer to one specific vocabulary, but rather to a general approach of using these attributes. Each application is assigned its own vocabulary and associated syntax.

- **Example:**

hCard is used to describe people and organizations, hCalendar to publish calendar data, and hAtom for news feeds. The community around the <http://www.microformats.org> site is set up to guide the process of developing and standardizing new Microformats. These Microformat vocabularies are rarely defined from scratch; instead, they are a translation of existing vocabularies to the Microformats approach

(e.g., hCalendar is a translation of vCalendar, hAtom of ATOM, hCard for vCard, etc.).

hcard and hcalendar



a. Some new microformats

- Semantic information itself is encoded in attribute values using simple strings.
- The definition of a new Microformat consists of defining these strings and giving a textual description as to their intended meaning.
- **For example:**

hCalendar specification specifies the meaning of “dtstart” (the starting date of a calendar event), “location” (the location of a calendar event), or “geo” (the latitude and longitude values of the location). The attribute of choice to encode these values is the @class attribute.

As a very simple example, the following HTML:

```
<div class="vevent">  
  <a href="http://2008.sxsw.com/interactive/" class="url">  
    <span class="summary">SXSW Interactive</span>  
  </a>  
</div>
```

Common problems and their solutions using Microformats + GRDDL

Problem	Solution in Microformat + GRDDL
Define terms in Microformats	Define attribute values, usually for <code>@class</code> , (sometimes for <code>@title</code> or the <code>@abbr</code> element)
Convert terms to RDF	Apply a GRDDL transformation (usually XSLT)
Find GRDDL transform (general)	Use <code>@profile</code> with generic value and a <code><link></code> element with <code>@rel="transformation"</code>
Find GRDDL transform (for specific, frequent cases)	Use <code>@profile</code> with a specific value (leading to another document to be GRDDL-d)

Extended by adding the start and end dates of the event

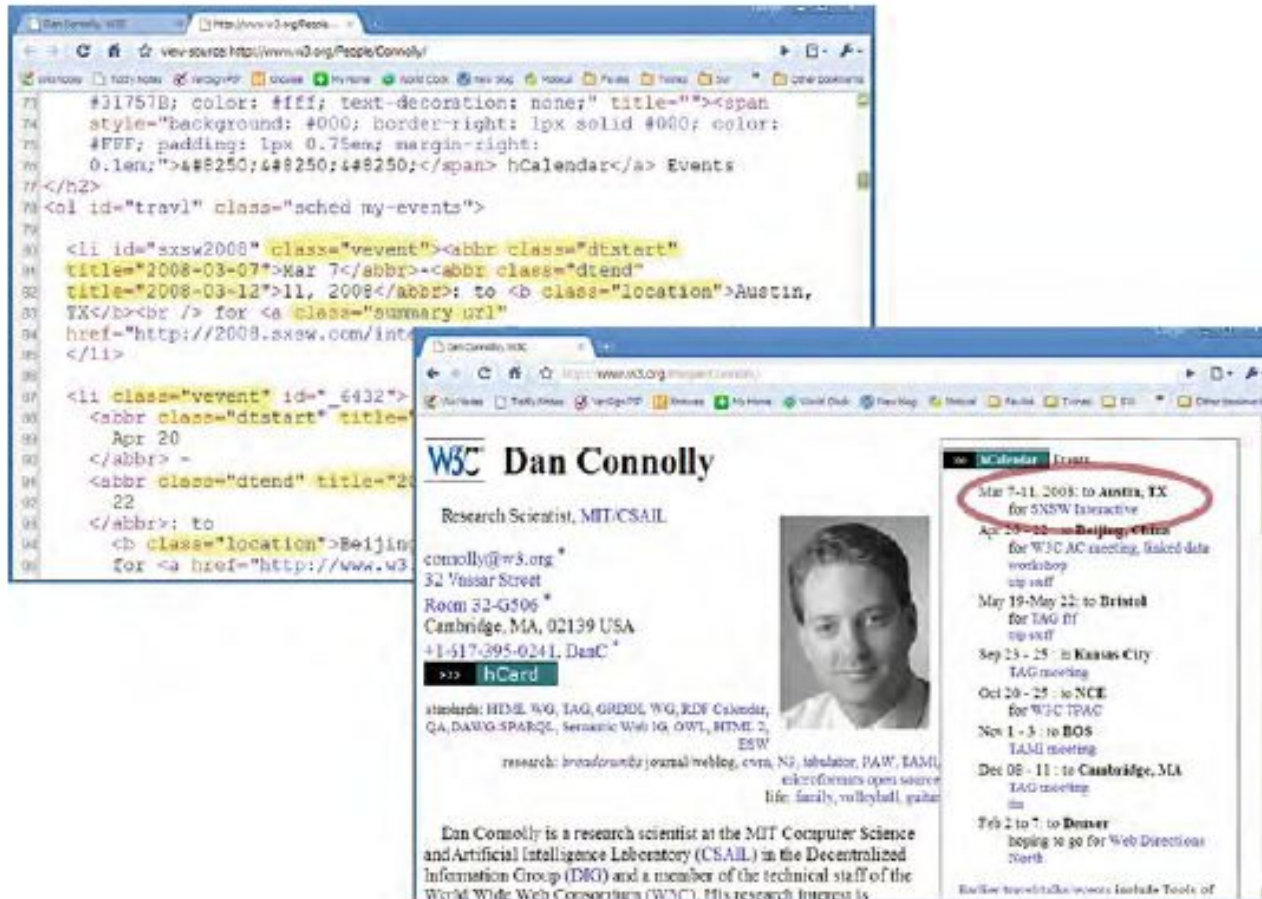
It is, in some cases, not possible to rely exclusively on @class and the existing HTML content structure; extra elements and possibly other attributes should also be used.

For example:

In hCalendar, is the setting of dates, which usually relies on the abbr element and uses the @title attribute for the precise (i.e., ISO formatted) date value instead of the human-readable form.

```
<div class="vevent">
  <abbr class="dtstart" title="2008-03-07">Mar 7</abbr>-
  <abbr class="dtend" title="2008-03-12">11, 2008</abbr>:
  <a href="http://2008.sxsw.com/interactive/" class="url">
    <span class="summary">SXSW Interactive</span>
  </a>
</div>
```


Typical example of hCalendar usage



- Page uses two different Microformats (hCard and hCalendar)
- Web page address data and conference events of the person.
- Extra information is embedded in the XHTML page, but those extra attributes are not visible on the browser screen.
- iCalendar data from a hCalendar Microformat markup.

**Extraction
Tool Building**

iCalendar

```
BEGIN:VCALENDAR
...
BEGIN:VEVENT
URL:http://2008.sxsw.com/interactive/
LOCATION;CHARSET=UTF-8:Austin\, TX
SUMMARY;CHARSET=UTF-8:SXSW Interactive
DTSTART:20080307
DTEND:20080312
END:VEVENT
```

b. Transforming Microformats to RDF

- If the host language is XHTML, then the straightforward way of doing that is to use a suitable XSLT transformation.
- XSLT is a functional style programming language, describes the transformation of an XML tree into either another XML tree or simply into plain text.
- Since its inception, XSLT has become an integral part of the XML infrastructure.
- Implementations of XSLT processors are widely available as stand-alone programs and as part of XML editors or Web browsers.
- Due to the simplicity of the usual Microformat dialects, the definition of an XSLT transformation to extract the semantic data and generate an RDF equivalent in RDF/XML (or Turtle) is usually a relatively straightforward task.

XHTML (A1[CSE344]--URI) □ microformats (processors -- Transformation) □ XML □
RDF □ Logic(OWL) □ Ontology □ web site (Resources) □ microformats (GRDDL/RDFa)
□ XML

XHTML (A2[CDE456] -- URI)

RDF/XML – Transformation [XML/RDF Processors]

```
<http://www.w3.org/People/Connolly/#sxsw2008>
  a <http://www.w3.org/2002/12/cal/icaltzd#Vevent>;
  <http://www.w3.org/2002/12/cal/icaltzd#summary>
    "SXSW Interactive";
  <http://www.w3.org/2002/12/cal/icaltzd#dtstart>
    "2008-03-07"^^xsd:date;
  <http://www.w3.org/2002/12/cal/icaltzd#dtend>
    "2008-03-12"^^xsd:date;
  <http://www.w3.org/2002/12/cal/icaltzd#url>
    <http://2008.sxsw.com/interactive/>;
  <http://www.w3.org/2002/12/cal/icaltzd#location>
    "Austin, TX".
```

Transform the Microformat data into RDF, integrating with other types of data on the Web

1. Define an RDF vocabulary corresponding to the Microformat dialect.
2. Create an XSLT transformation that extracts the data from the XHTML source and produces RDF using that vocabulary.

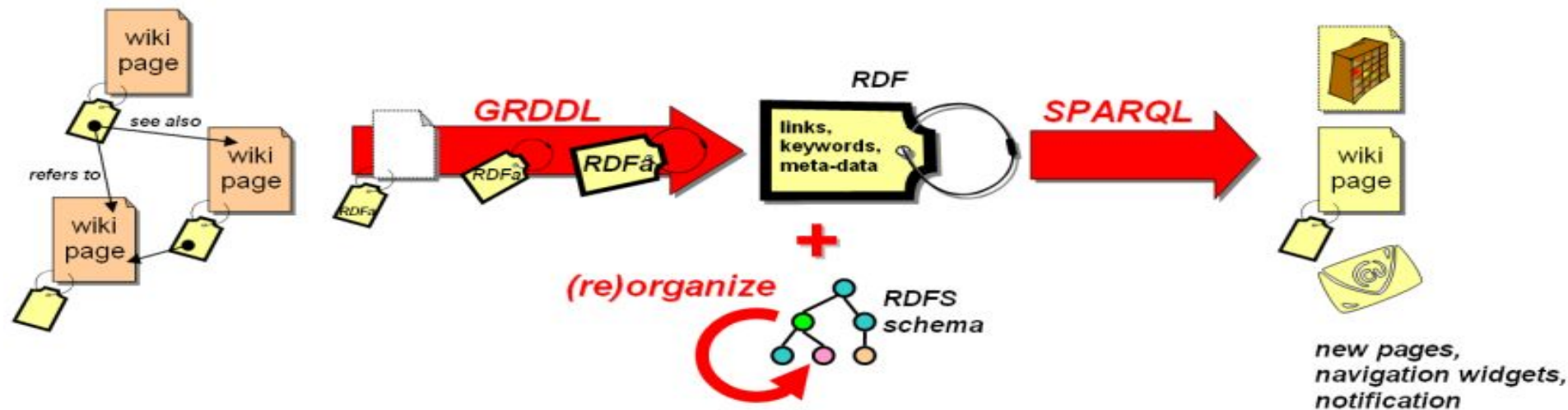
<https://www.w3.org/wiki/CustomRdfDialects>

HTML-based Dialects

Dialect	Domain	Schema	Mapping	Status
Dublin Core in HTML (legacy)	Bibliographic Metadata	Dublin Core	dc-extract	Valid 2003-2003, now superseded
Dublin Core in HTML (current)	Bibliographic Metadata	Dublin Core	dc-html2rdfxml.xsl	Maintained by DCMI
eRDF	generic	RDF	extract-rdf.xsl , profile	by Ian Davis
GeoURL markup	Geo (lat/long)	GeoInfo	grokGeoURL.xsl	e.g. Joe Lambda's homepage
glink	HTML links/link@rel	#html, DC	groklinks.xsl	maintained by Danny Ayers
hCalendar	Calendars and Events	RDF Calendar	glean-hcal.xsl , hcal profile	maintained by Dan Connolly since Dec 2002
hCard	Contact Information	vcard ontologies (2006)	hcard2rdf.xsl , hcard profile	maintained by Walsh , Halpin , and Suda since Nov 2006; used in xtech schedule
hDoap	Software project description	DOAP	hDoap	maintained by Danny Ayers
hReview	Opinions, Ratings and Reviews	RDF Review	hreview2rdfxml.xsl	See RDF Review
hTurtle	generic	RDF	XSLT service , profile	by Sean Palmer
metaprof	meta/link@rel + generic	various	xh2rdf.xsl	maintained by Masahide Kanzaki
RDFa	generic	RDF	RDFa profile	by Fabien Gandon , collaborating with the GRDDL and RDFa WGs

C. GRDDL

- GRDDL uses existing (X)HTML constructs to locate the necessary transformation(s) for the HTML markup.
- Generic GRDDL processor would then find and download the transformation (at least conceptually, because an optimized GRDDL processor would probably cache the transformation for all major Microformats), execute the transformation, and produce the RDF output.
- Fully generic and automatic GRDDL processors can be created and added to general RDF application environments.



Way for GRDDL Transformation

The simplest way of defining the GRDDL transformation is to modify the header of an XHTML file as follows:

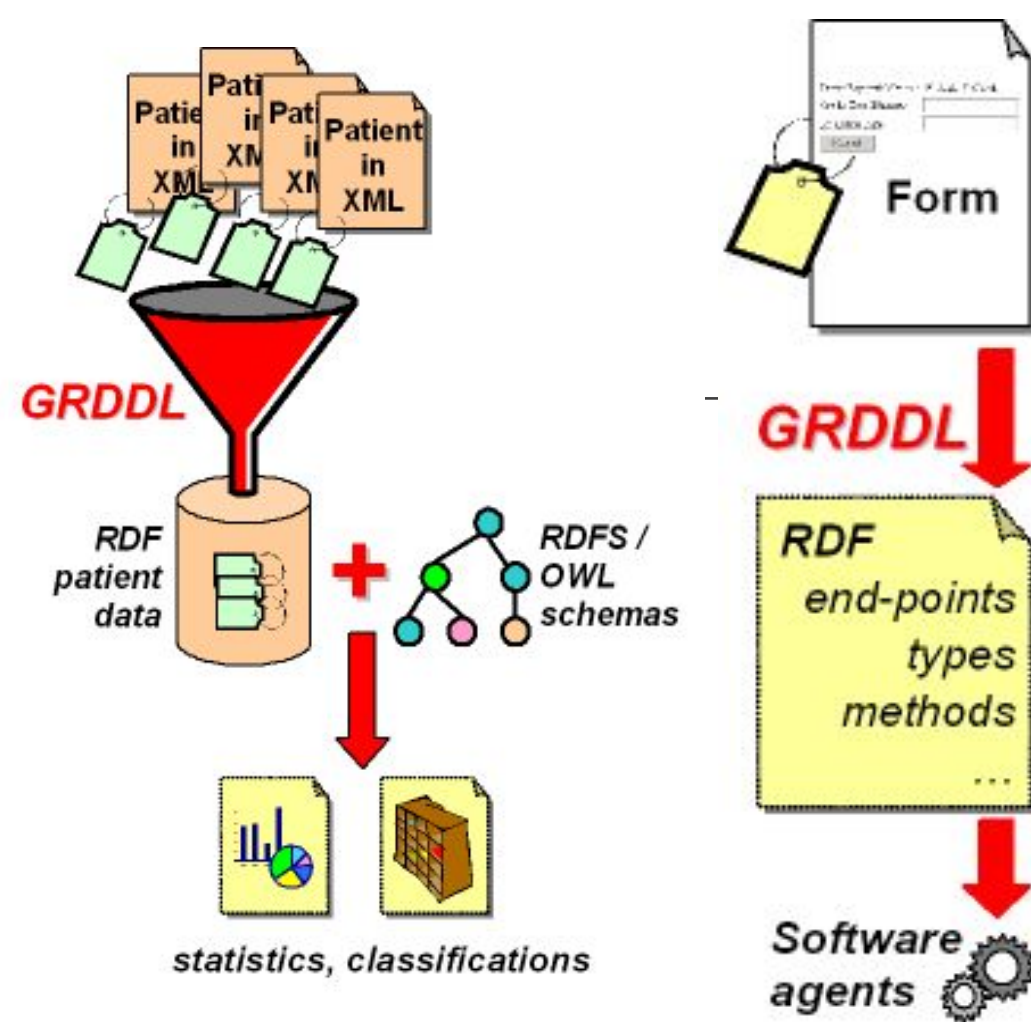
- Add a @profile attribute to the XHTML head denoting the fact that the XHTML file is to be processed by a GRDDL processor; and use a “link” element in the head to indicate the URI of the transformation.

```
<html>
  <head profile="http://www.w3.org/2003/g/data-view">
    <link rel="transformation"
      href="http://www.w3.org/2002/12/cal/glean-hcal.xsl"/>
    ...
  </head>
  ...
</html>
```

- If the page uses several Microformats, separate link elements should be added for each of those.
- The GRDDL processor would then execute each transformation separately and perform an RDF merge on the generated graphs, yielding the final result.

GRDDL process

- GRDDL defines a way to add the necessary attributes to a dialect of XHTML files, simplifying the task of authors.
- The profile value <http://www.w3.org/2003/g/data-view> simply means “process this file through a GRDDL processor,” publishers can also use dialect-specific @profile values.
- The goal of these specific values is to “merge” the two items of information above, that is, making the separate link element unnecessary.



1. The GRDDL processor follows this special profile URI, retrieves the target (i.e., profile) document.
2. Performs a GRDDL transform on the profile document, yielding a (small) RDF graph.
3. This RDF graph contains a reference to another transformation.
4. This transformation is applied on the original source.